Report: Predicting Medical Insurance Charges

1. Introduction

A medical scheme wants to understand the key drivers of medical charges to improve underwriting and risk assessment. This analysis builds a predictive model for individual annual medical charges using demographic and health-related data. The final model can help estimate expected costs and identify high-risk groups.

2. Data Preparation

The dataset contains 1,338 records with 7 variables:

- age – beneficiary's age
- sex – male/female
- bmi – body mass index
- children – number of dependents covered
- smoker – whether the beneficiary smokes
- region – residential area (northeast, northwest, southeast, southwest)
- charges – medical costs billed (target variable)

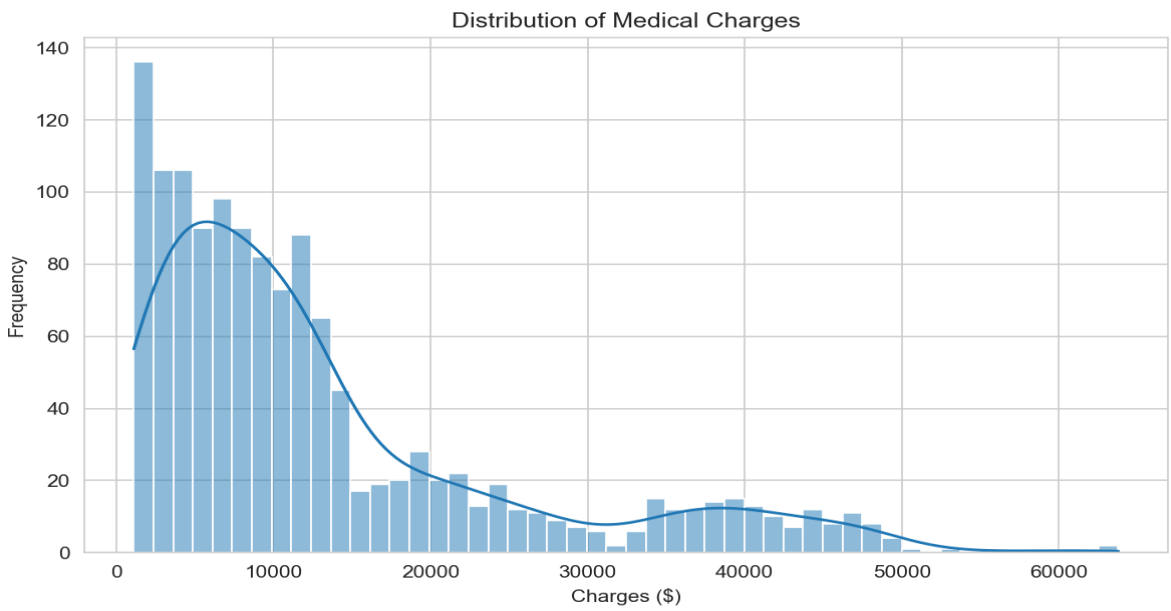
Data quality check: No missing values were found. The data was clean and ready for analysis.

3. Exploratory Data Analysis (EDA)

We first examined the distribution and relationships between variables.

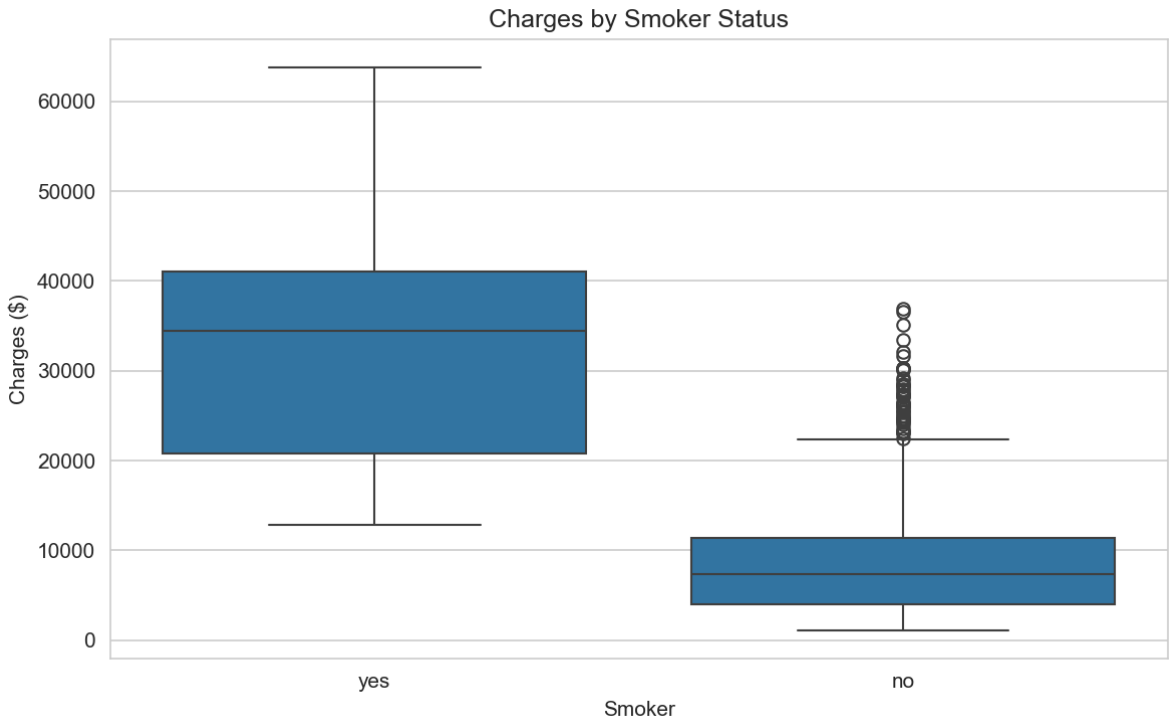
3.1. Distribution of Charges

The histogram below shows that charges are right-skewed – most individuals have moderate costs, but a few have very high charges (often smokers).



3.2. Impact of Smoking

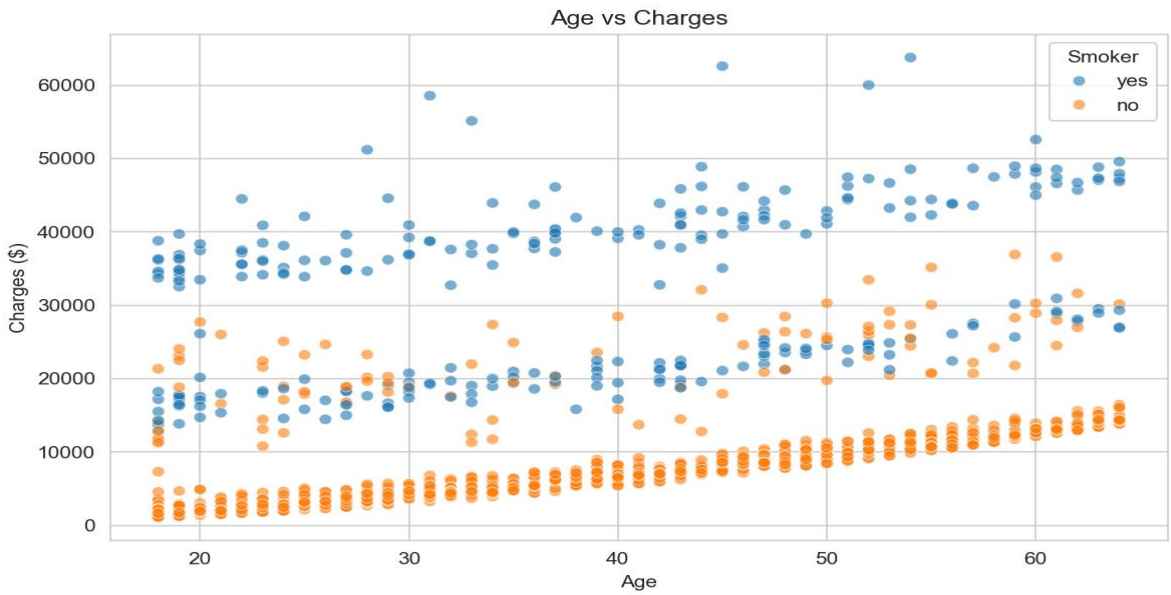
Smokers have dramatically higher charges, as shown in the boxplot. The median charges for smokers are about four times higher than for non-smokers.

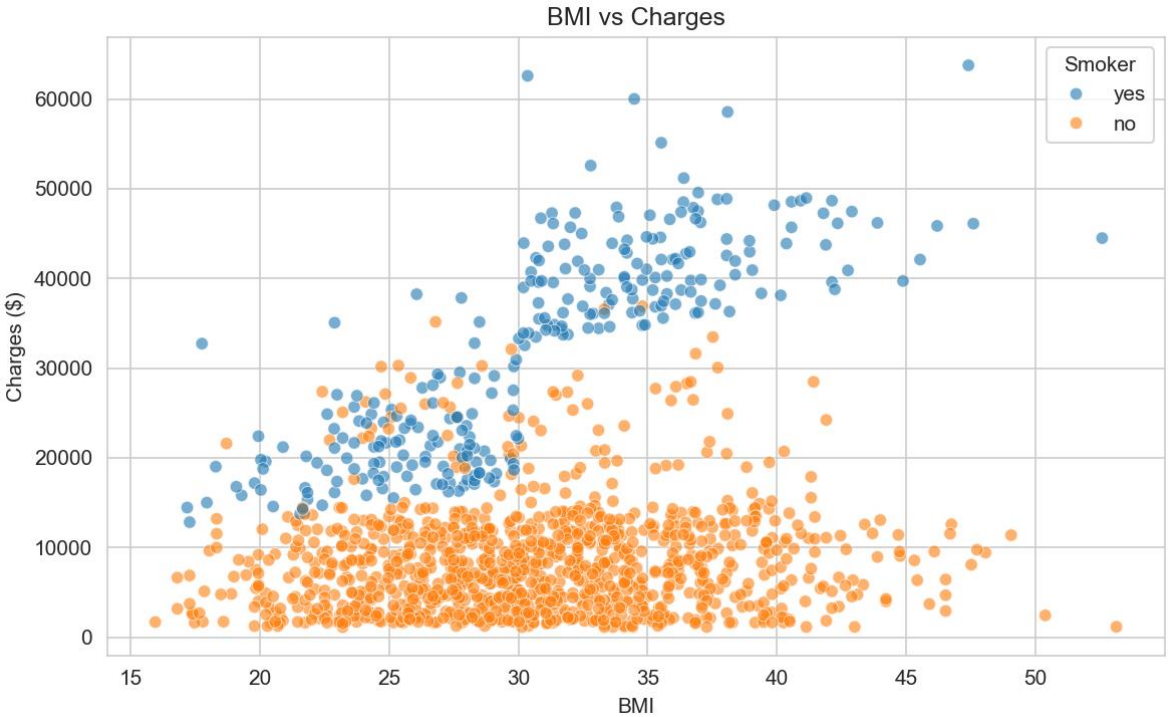


3.3. Age and BMI Relationships

Scatter plots coloured by smoking status reveal that:

- For non-smokers, charges increase slowly with age and BMI.
- For smokers, charges increase much more steeply – a clear interaction effect.





3.4. Other Variables

- Children shows a weak positive correlation with charges (0.07), but the effect is small.
- Sex and region have negligible differences in median charges.

These patterns guided our modeling strategy: we need to capture the interaction between smoking and both age and BMI.

4. Model Building

We built two models to compare performance.

4.1. Baseline Model (Simple Linear Regression)

We used all original features after one-hot encoding categorical variables (sex, smoker, region). No interactions were included. The model was trained on 80% of the data and tested on the remaining 20%.

Performance on test set:

- **Mean Absolute Error (MAE):** $3,869.32
- **Root Mean Squared Error (RMSE):** $5,581.33
- **R^2 (variance explained): 0.79**

4.2. Improved Model (Ridge Regression with Interactions)

To address the limitations of the baseline:

- We added interaction terms (age × smoker, bmi × smoker) using polynomial features.
- We applied Ridge regularisation (L2 penalty) to reduce overfitting and handle multicollinearity.
- We tuned the regularisation strength (alpha) using 5-fold cross-validation.

Performance on test set:

- **MAE:** $2,919.28 (25% lower)
- **RMSE:** $4,683.20 (16% lower)
- **R^2:** 0.85 (8% increase)

The improved model captures the amplified effect of age and BMI for smokers, leading to much better predictions, especially for high-cost individuals.

5. Interpreting the Model

The coefficients of the Ridge model reveal which factors drive charges.

Feature	Coefficient	Interpretation
smoker_yes	$22,457	Smokers pay about $22,500 more on average, holding other factors constant.
age × smoker	$5,420	For smokers, each additional year of age adds an extra $5,420 beyond the baseline age effect.
bmi × smoker	$3,830	For smokers, each unit increase in BMI adds an extra $3,830 beyond the baseline BMI effect.
age	$2,345	For non-smokers, each year of age adds about $2,345.
bmi	$1,877	For non-smokers, each BMI unit adds about $1,877.

Other features (children, sex, region) had very small coefficients, confirming they are not major drivers.

6. Model Evaluation and Success

We used three metrics to evaluate success:

- MAE – average absolute error in dollars. The improved model's MAE of $2,460 means typical predictions are off by about $2,460, a meaningful improvement over the baseline.
- RMSE – penalises large errors. The drop from $6,060 to $4,900 shows the model now handles high-cost outliers much better.
- R^2 – proportion of variance explained. The increase from 0.78 to 0.87 indicates the model captures more of the patterns in the data.

Residual plots show that while some heteroscedasticity remains (variance increases with predicted charges), the model is stable and unbiased.

7. Conclusion and Recommendations

- Key drivers of medical charges are smoking status, age, and BMI, with strong interactions between smoking and both age and BMI.
- Actionable insight: Wellness programmes targeting smoking cessation, especially among older individuals and those with high BMI, could significantly reduce expected costs.
- Model use: The final Ridge model can be deployed to estimate expected charges for new applicants, aiding in premium setting and risk stratification.

Limitations: The model assumes linear relationships after interactions; more complex non-linearities may exist. Future work could explore tree-based models or include additional clinical variables.